

# A 0D pulsatile model of aortic-arch banding

Reproducing carotid pulsatility redistribution (Eberth et al., 2009)

svZeroDSolver

July 25, 2026

# The problem

**Eberth et al. (2009):** a stiff band on the mouse aortic arch, placed *between* the two carotid takeoffs, creates paired carotids with **similar mean pressure & flow but very different pulsatility**.

- **RCCA-B** (right) — *upstream* of the band: sees the full pulse + reflections  $\Rightarrow$  pulsatility index rises ( $PI \approx 3.11$ ).
- **LCCA-B** (left) — *downstream*: the resistive–inertial band damps the pulse ( $PI \approx 1.65$ ); baseline CCA  $\approx 1.16$ .

**Goal:** a reduced-order (0D) lumped model that reproduces this redistribution with the band as the only structural change — and that maps to the wall growth & remodeling Eberth measured.

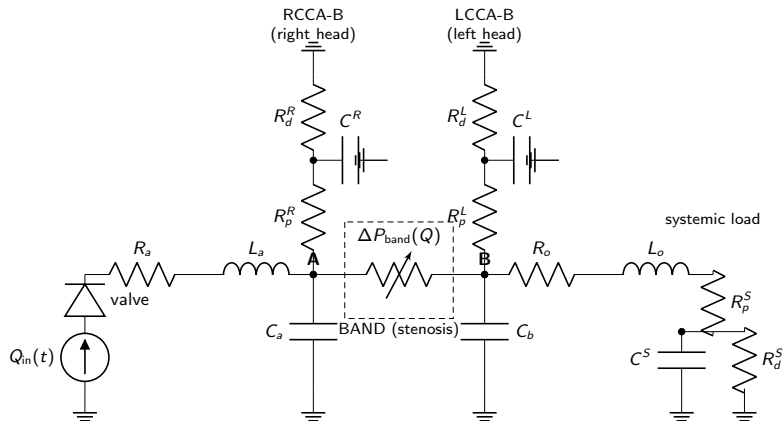
# The 0D model

Heart  $\rightarrow$  aortic valve  $\rightarrow$  ascending aorta ( $R_a, L_a, C_a$ )  $\rightarrow$  **node A**  $\rightarrow$  band  $\Delta P_{\text{band}}(Q) \rightarrow$  **node B**  $\rightarrow$  descending aorta ( $R_o, L_o, C_b$ ); a carotid RCR bed ( $R_p, C, R_d$ ) hangs off each node.

- Vessels: RCL BloodVessel blocks
- Band: nonlinear *Young-Tsai* stenosis
$$\Delta P_{\text{band}} = (R_{\text{band}} + S|Q|)Q + L_{\text{band}} \dot{Q}$$
- Terminal beds: 3-element RCR windkessels
- Valve: ValveTanh diode
- Parameters from mouse literature (geometry, blood, wall stiffness  $E$ )
- Every parameter tagged  
FIXED / DERIVED / CALIBRATE
- Only the band `stenosis_coefficient` is calibrated (to the Table-1 PI ratio)

Three states differ *only* in carotid geometry: pre-banding (baseline), acute (baseline walls), chronic (Table-1 remodeled walls).

# 0D circuit (electric analogy)

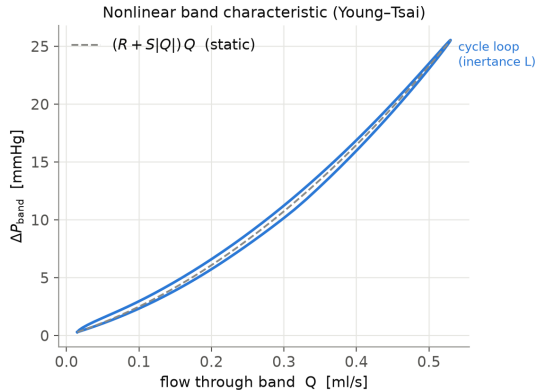


| symbol               | component (this repo) |
|----------------------|-----------------------|
| $Q_{in}(t)$          | INFLOW                |
| valve                | aortic_valve          |
| $R_a, L_a, C_a$      | ascending_aorta       |
| $\Delta P_{band}(Q)$ | aortic_band           |
| $R_o, L_o, C_b$      | descending_aorta      |
| node A / B           | node_A / node_B       |
| $R_p^R, C^R, R_d^R$  | RCR_RIGHT             |
| $R_p^L, C^L, R_d^L$  | RCR_LEFT              |
| $R_p^S, C^S, R_d^S$  | RCR_SYS               |

Not drawn (negligible): aortic\_root ( $R, L$  before the valve) and the carotid conduits `rcca.b/lcca.b` ( $R, L, C$  between node A/B and `RCR_RIGHT/RCR_LEFT`) — their own wall compliance is tiny next to the bed/aortic  $C$ .

Pressure  $\leftrightarrow$  voltage, flow  $\leftrightarrow$  current. Node A (RCCA-B) is upstream of the band and stays pulsatile; node B (LCCA-B) is downstream and damped. Each terminal bed is an  $R_p$ - $C$ - $R_d$  Windkessel.

# The band is a nonlinear (Young–Tsai) element



$\Delta P$ – $Q$  across the band: the quadratic Bernoulli/turbulent term ( $S|Q|Q$ ) plus an inertance loop ( $L \dot{Q}$ ). This nonlinearity + inertia is what damps the downstream pulse while leaving the upstream node pulsatile.

# Parameter roles: fixed, assumed, and the knob we tune

## Fixed (measured / literature)

- blood  $\rho = 1.06$ ,  $\mu = 3.5$  cP
- wall  $E$ : aorta 1.0, carotid 1.5 MPa
- HR 6.09 Hz, CO 0.20 ml/s, MAP 90
- carotid ID/wall (Table 1), aorta dia.
- measured central aortic compliance

⇒ all vessel  $R$ ,  $L$ ,  $C$  are *derived* from these via Eq. (1)–(3).

## Assumed (plausible, not fitted)

- segment lengths (carotid, band) — estimates
- ejection waveform shape (systolic fraction)
- RCR split  $R_p:R_d = 10:90$ ; time const.  $\tau$
- distal pressure  $P_d = 0$
- valve  $R_{max}/R_{min}$ /steepness

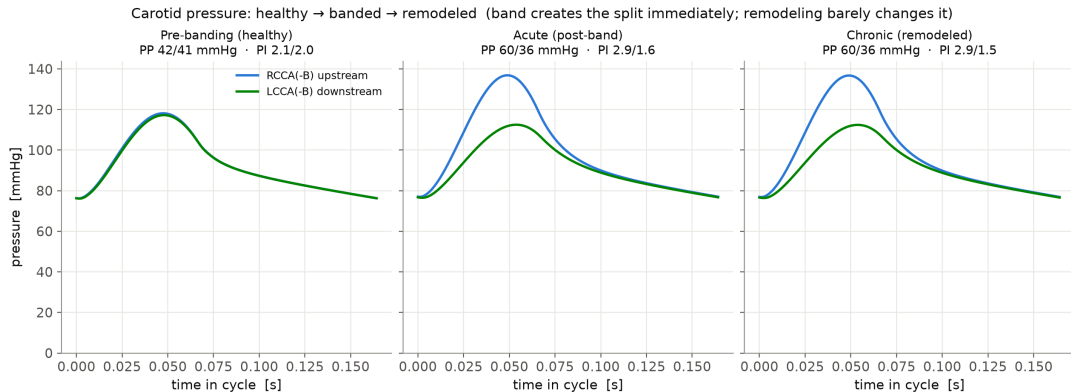
chosen by convention/estimate — could be refined, but not tuned to the data.

## Knob (calibrated to data)

- band stenosis\_coefficient  $S$
- the **one** parameter we turn
- geometry gives a prior; tuned to the Table-1 PI ratio (3.11/1.65)

Everything else is fixed or derived — this is the only free knob fitted to the data.

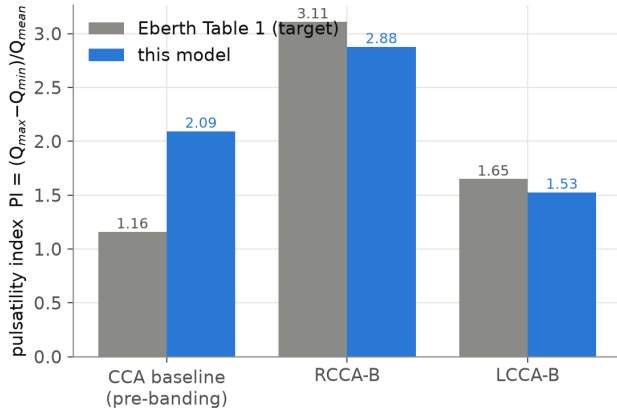
# Result: banding creates the split immediately



Carotid pressure is **symmetric before banding**, splits **immediately** on banding (upstream  $\uparrow$ , downstream damped), and **barely changes** with wall remodeling  $\Rightarrow$  the split is set by band *position*; remodeling is the response, not the cause.

# Result: pulsatility vs. the paper

Pulsatility by location — calibrated model vs paper Table 1  
(stenosis calibrated to the PI ratio; both PIs within ~7%)



|           | Paper   | Model         |
|-----------|---------|---------------|
| RCCA-B PI | 3.11    | $\approx 2.9$ |
| LCCA-B PI | 1.65    | $\approx 1.5$ |
| PI ratio  | 1.89    | 1.89          |
| PP RCCA-B | 56      | 60            |
| MAP (R/L) | similar | 98/91         |

The **banded** carotids, the PI ratio, MAP similarity, and pulse pressure are reproduced; mean carotid flows land in the measured range.



## Limitation: baseline vs. banded cannot both be matched in 0D

- The model **over-predicts the baseline (pre-banding) PI** ( $\approx 2.1$  vs. the paper's 1.16), though baseline *pulse pressure* matches ( $\approx 41$  vs. 42).
- A parameter sweep shows every knob that lowers the baseline PI to 1.16 **also collapses the banded PI** — the passive bed is a linear pulsatility filter.
- The paper's baseline→banded amplification is  $\sim 2.7\times$  ( $1.16\rightarrow 3.11$ ); a lumped 0D band delivers only  $\sim 1.5\times$ .
- The missing  $\sim 2\times$  is **wave reflection** off the band — a spatial/timing effect a 0D element cannot represent.

⇒ Reproducing the *full* amplitude needs a **1D** (wave-propagation) model — exactly the 0D→1D boundary.

# Targets: pinned, fixed, or an actual free knob?

## Control group (CCA) — 1 free knob

| target          | how it's matched  | status      |
|-----------------|-------------------|-------------|
| $\bar{Q}=0.016$ | bed $R_p+R_d$     | pinned      |
| MAP=92          | sys $R_p^S+R_d^S$ | pinned      |
| PI=1.16         | bed $C^{R,L}$     | <b>KNOB</b> |

## Banded group (RCCA-B / LCCA-B) — 3 free knobs

| target                | how it's matched                  | status      |
|-----------------------|-----------------------------------|-------------|
| $\bar{Q}=0.022/0.012$ | bed $R_p^R+R_d^R$ / $R_p^L+R_d^L$ | pinned      |
| MAP=86/77             | sys $R$ (pinned) + band $S$       | <b>KNOB</b> |
| PI=3.11/1.65          | bed $C^R$ / $C^L$                 | <b>KNOB</b> |

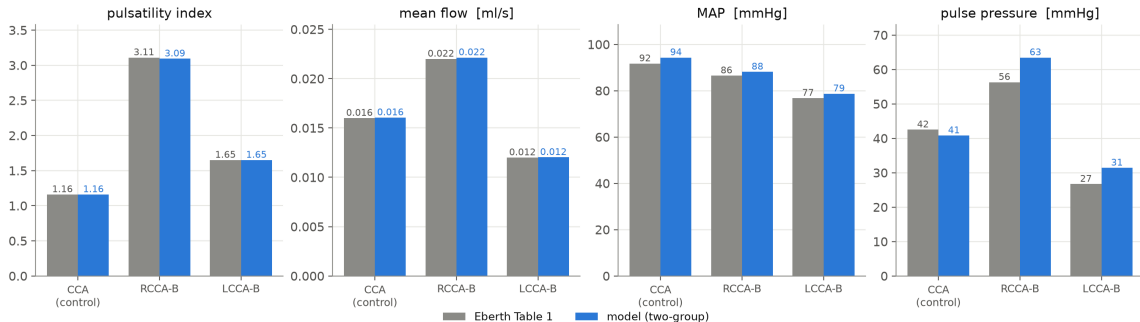
**Only 4 parameters are actually free/calibrated** (control bed  $C^{R,L}$ ; banded bed  $C^R, C^L$ , band  $S$ ). Bed resistances ( $R_p+R_d$ , all beds) are *pinned* — solved directly from  $(\bar{Q}, \text{MAP})$  via Ohm's law, not searched. (PP omitted above: it's matched by the aortic  $C_a, C_b$ , which is *fixed* literature identical in both groups — an *emergent* match, not a tuned one.)

Also fixed: HR (7.17/6.09), CO, carotid geometry, blood  $\mu, \rho$ , wall moduli  $E$ , ejection shape.

# Revised: two-group calibration (control vs. banded)

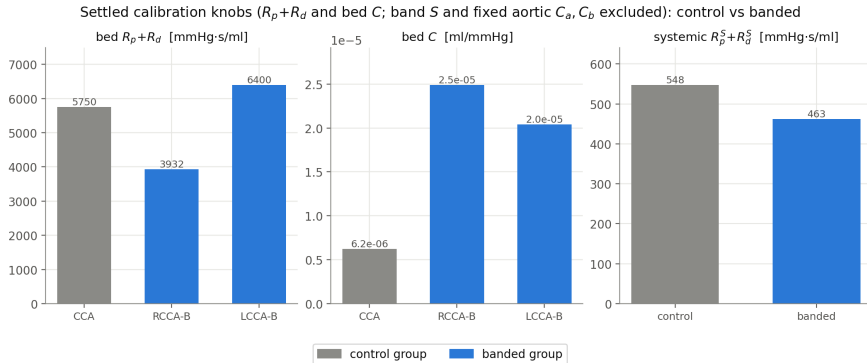
Control (CCA) and banded (RCCA-B/LCCA-B) are **two separate animal groups** — they need not share  $R/C/L$ . Calibrating each on its own terms — HR and measured flows/pressures pin the beds, the band anchored to the measured  $A \rightarrow B$  pressure drop, per-carotid bed compliance sets the PI — matches *all three* carotid states with a stock RCR:

Two-group calibration vs Eberth Table 1 (control CCA + banded RCCA-B/LCCA-B)



Only **4 parameters are free** (bed  $C^{R,L}$ ,  $C^R$ ,  $C^L$ , band  $S$  — previous slide); bed  $R$  pinned, aortic  $C$  fixed. Cost: bed compliance differs between groups (fitted) — descriptive, not predictive.

# Settled calibration knobs: control vs. banded



$R_p+R_d$  (bed  $R$ ) sets the flow split (RCCA-B low  $\rightarrow$  more flow); the banded bed *compliance*  $C^R, C^L$  is  $\sim 3\text{--}4\times$  the control's  $C^{R,L}$  — a **fitted** difference, not measured (opposite to what autoregulation would do). Systemic  $R_p^S+R_d^S$  is essentially unchanged. (Aortic  $C_a, C_b$  omitted here — it's fixed literature, identical in both groups, not a calibrated knob.)

# Summary

- A literature-parameterized 0D model reproduces the Eberth **pulsatility redistribution**: upstream carotid pulsatile, downstream damped, at matched mean pressure/flow.
- Banding creates the split **immediately**; wall remodeling barely changes it — consistent with pulsatility being the *stimulus* for remodeling.
- With a *single shared* parameter set, 0D matches the redistribution and pressures but not both the baseline and banded PIs (a bounded amplification limit; a 1D wave-reflection effect is small for the mouse).
- Treating control and banded as *separate groups* — their own beds, band anchored to the measured pressure drop — matches **all three carotid states**, at the cost of group-specific (fitted) bed compliance.